

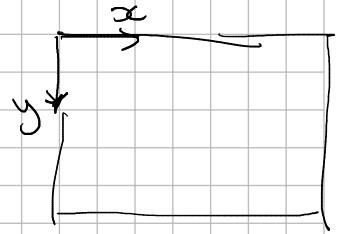
$$\underline{u} = \begin{bmatrix} u \\ v \\ w \end{bmatrix} = \begin{bmatrix} u(x,y,t) \\ v(x,y,t) \\ 0 \end{bmatrix}$$

Peguen deformaciones

$$u = A(e^{\alpha ky} + \gamma \cdot e^{\beta ky}) \cdot \cos[k(x - ct)]$$

$$v = A(\alpha e^{\alpha ky} + \beta \cdot e^{\beta ky}) \cdot \cos[k(x - ct)]$$

$$w = 0$$



$$\alpha = -0,8475$$

$$\beta \cdot \gamma + \beta + 2\alpha = 0$$

$$\beta = -0,3933$$

$$\beta \cdot \beta - \gamma = 0$$

$$\gamma = -0,15773$$

$$\beta = 1,4679$$

$$\epsilon_{ij} = \frac{1}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right)$$

$$\epsilon_{zz} = \frac{\partial u_z}{\partial z} \Rightarrow \partial u_z = \int \epsilon_{zz} \cdot dz$$

deformación unitaria

$$u_z(z) - u_z(z_0) = \epsilon_{zz} \cdot (z - z_0)$$

$$\epsilon_{zz} = \frac{\Delta u_z}{(z - z_0)} = \frac{\Delta u_z}{L}$$

$$\epsilon_{xx} = \frac{\partial u_x}{\partial x} = \frac{\partial u}{\partial x} = A(e^{\alpha ky} + \gamma \cdot e^{\beta ky}) \cdot -\sin[k(x - ct)] \cdot k$$

$$= -Ak(e^{\alpha ky} + \gamma \cdot e^{\beta ky}) \cdot \sin[k(x - ct)]$$

$$\epsilon_{yy} = \frac{\partial u_y}{\partial y} = \frac{\partial v}{\partial y} = Ak(e^{\alpha ky} \cdot \alpha + e^{\beta ky} \cdot \beta) \cdot \cos[k(x - ct)]$$

$$\epsilon_{xy} = \frac{1}{2} \left(\frac{\partial u_x}{\partial y} + \frac{\partial u_y}{\partial x} \right) = A(e^{\alpha ky} - e^{\beta ky}) \cdot k \alpha \cdot \cos[k(x - ct)]$$

$$\left. \begin{matrix} \epsilon_{xx} \\ \epsilon_{yy} \\ \epsilon_{xy} \end{matrix} \right\} \rightarrow \epsilon_{mox} \rightarrow \frac{\partial \epsilon_{mox}}{\partial x} = 0$$

$$\frac{\partial \epsilon_{min}}{\partial y} = 0$$

$$\sigma_{max} = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2} \right)^2 + \tau_{xy}^2}$$

$$\sigma_{min}$$

$$\left. \begin{matrix} \epsilon_{xx} \\ \epsilon_{yy} \\ \epsilon_{xy} \end{matrix} \right\} \rightarrow \sigma_x = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2} \cos 2\theta + \tau_{xy} \sin 2\theta$$

$$\left\{ \begin{matrix} \frac{\partial \epsilon_{x'x'}}{\partial \theta} = 0 \\ \frac{\partial \epsilon_{x'x'}(x')}{\partial x} = 0 \\ \frac{\partial \epsilon_{x'x'}(y')}{\partial y} = 0 \end{matrix} \right.$$

Al ser una función exponencial (de exponente negativo) el valor de las deformaciones disminuye con los ~~coordenadas~~ ~~coordenadas~~ y.

$$y=0$$

$$\epsilon_{xy} = A \cdot \left(e^{k \cdot 0 \cdot x} - e^{k \cdot 0 \cdot y} \right) \cdot k \cdot \alpha \cdot \cos \left[k(x - t \cdot C_R) \right] = 0 \Rightarrow \text{las deformaciones máximas en } x \text{ e } y \text{ son las máximas}$$

$$\begin{cases} \epsilon_{xx} = -A k (1 + \gamma) \cdot \sin \left[k(x - t \cdot C_R) \right] \\ \epsilon_{yy} = A k (\alpha^2 + \gamma) \cdot \sin \left[k(x - t \cdot C_R) \right] \end{cases}$$

$$\begin{aligned} \epsilon_{xx} &= \mp A k (1 + \gamma) = \mp 0,4227 A k \\ \epsilon_{yy} &= \pm A k (\alpha^2 + \gamma) = \pm 0,1410 A k \end{aligned} \quad \left\{ \begin{aligned} \epsilon_{min} &= -0,4227 A k, \quad \epsilon_{max} = 0,1410 A k \\ \epsilon_{min} &= -0,1410 A k, \quad \epsilon_{max} = 0,4227 A k \end{aligned} \right.$$

Para la gráfica pueden usar:

$$A = 0,4$$

$$k = 2$$

$$C_R = 1$$